

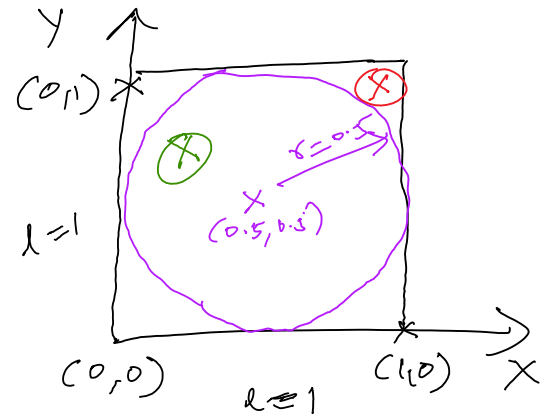
$$\pi = 3.14 \dots$$

Pi Estimator Project (#4) using OpenMP Multithreading

$$r = 0.5$$

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$$\begin{aligned} & \text{ratio of } \frac{\text{area of the circle}}{\text{area of the square}} \\ &= \frac{\pi r^2}{l^2} = \frac{\pi (0.5)^2}{1^2} \\ &= \frac{\pi}{4} \end{aligned}$$



Estimate Pi (int n: #darts thrown)

```
{
  int hits = 0;
  int i = 0;
  for (i = 0; i < n; i++)
```

{

a) throw a random dart $\rightarrow \langle x, y \rangle$

b) check if $\langle x, y \rangle$ is inside the circle)

yes

hits++

}

$$\text{Prob (dart landing inside the circle)} = \frac{\text{Area of circle}}{\text{Area of the square}}$$

$$\text{hit rate} = \frac{\# \text{hits}}{n} \approx \frac{\pi}{4}$$

$$\Rightarrow \pi = \left(\frac{\# \text{hits}}{n} \right) \times 4 \approx \text{estimate for } \pi$$

dart
 (X) dart lands outside the circle
 "hit" $\Leftarrow$ (X) dart lands inside the circle

Experiments

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correct = 3.141592653589793

precision testing

keep $\frac{n}{p}$ fixed.

$p=1 \dots 8 \rightarrow$

Performance Testing

